Fractions and Binomial

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The binomial coefficient, $\binom{n}{k}$, is defined by the expression:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Fractions can be used inline within the paragraph text, for example $\frac{1}{2}$, or displayed on their own line, such as this:

$$\frac{1}{2}$$

We use the `amsmath` package command `\text{...}` to create text-only fractions like this:

$$\frac{\text{numerator}}{\text{denominator}}$$

Without the `\text{...}` command the result looks like this:

$$\frac{numerator}{denominator}$$

Fractions typeset within a paragraph typically look like this: $\frac{3x}{2}$. You can force LaTeX to use the larger display style, such as $\displaystyle\frac{3x}{2}$, which also has an effect on line spacing. The size of maths in a paragraph can also be reduced: $\scriptstyle\frac{3x}{2}$ or $\scriptscriptstyle\frac{3x}{2}$. For the `\scriptscriptstyle` example note the reduction in spacing: characters are moved closer to the *vinculum* (the line separating numerator and denominator).

Equally, you can change the style of mathematics normally typeset in display style:

$$f(x)=\frac{P(x)}{Q(x)} \quad \text{and} \quad \textstyle f(x)=\frac{P(x)}{Q(x)} \quad \text{and} \quad \scriptstyle f(x)=\frac{P(x)}{Q(x)}$$